Lecture 1-1: JavaScript Introduction

What is JavaScript?

JavaScript (JS) is a **scripting language** used to make web pages interactive. It runs inside a **JavaScript engine** (for example, in the browser: V8/SpiderMonkey/WebKit JS engine). In your lecture file, you used JS inside a `<script>` tag.

How JavaScript Runs (Important Concept)

Unlike languages such as C/C++ that usually go through a compile step before execution, JavaScript code is executed by an engine at runtime. This is why many errors appear **when your code runs**.

> **Key takeaway:** JavaScript errors are typically runtime errors. Use the browser developer tools to debug.

Where Do We Write JavaScript in a Web Page?

- **Inline:** On HTML elements (not recommended for large code).
- **Internal:** In a `<script>` tag inside the HTML file.
- **External:** In a separate `.js` file linked using `<script src="...">`.

Developer Console Basics

In your lecture you used these common tools:

Method	What it does	Typical use
`console.log()`	Prints normal output	Debugging values
`console.warn()`	Prints a warning	Non-critical issues
`console.error()`	Prints an error	Invalid input / failures

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First JS Example (From Your Lecture)

```
alert("Hello");
console.log("Hello");
console.warn("A warning");
console.error("An error");

document.write("Hello from JS");
```

What to Notice

- JS is **case sensitive** (e.g., `Console` is not the same as `console`).
- Statements often end with `;`. JS can insert semicolons automatically, but it's safer to write them.
- `document.write()` works, but in real websites it's rarely used because it can replace the whole page if executed late.

Try It Yourself

Open a browser, press `F12` and go to the **Console** tab. Paste this:

```
console.log("JS is running!");
console.log("2 + 2 =", 2 + 2);
```

Output:

JS is running!
2 + 2 = 4

Best Practice Notes (Instructor Insights)

- Prefer `let` / `const` for variables in modern JS; `var` has confusing scoping rules.
- Use the console for debugging instead of guessing.
- Write small code, run often, fix errors early.

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